

Abstract

Modern warehouse management systems and automated logistics frameworks are predominantly reliant on centralized server architectures. This structural paradigm introduces a critical vulnerability: a single point of failure where the collapse of the central coordinator halts the entire automation matrix. This comprehensive report details the engineering, theoretical modeling, and practical implementation of a decentralized autonomous swarm robotics system optimized for automated warehouse material handling operations. By distributing computational intelligence across independent edge-computing nodes, the robotic swarm leverages a shared global environment visualization state to autonomously execute localization, path estimation, behavioral logic adjustments, and conflict resolution.

The system integrates advanced computer vision utilizing ArUco fiducial markers for digital twin synthesis, differential drive kinematics for precise mechatronic navigation, and specialized scissor-linkage end-effectors analyzed via the principle of virtual work. Furthermore, the report addresses critical networking challenges inherent in decentralized User Datagram Protocol (UDP) broadcasts across embedded microcontrollers, offering robust hardware and firmware optimizations to ensure seamless scalability and operational continuity. The structural engineering of the swarm agent is systematically designed to isolate heavy electromechanical assemblies from computational logic elements, mitigating electromagnetic interference and vibration harmonics. Through the application of Direct Linear Transform (DLT) algorithms for homography estimation, the system bridges the physical test track to virtual software frames, enabling real-time, millimeter-accurate coordinate scaling. This document serves as an exhaustive technical dossier, chronicling the integration of mechanical, electrical, and software domains to validate the efficacy of decentralized swarm logistics.

CHAPTER 1

1.1 Introduction & Project Architecture

This project details the engineering phase 1 implementation of a decentralized autonomous swarm robotics system optimized for automated warehouse material handling operations. Modern warehouse management systems typically rely on centralized server frameworks, creating a fatal single point of failure. If the central coordinator fails, the entire automation matrix collapses. This project counteracts that structural limitation by introducing a robust decentralized architecture.

1.2 Decentralized Swarm Principles

In this system, individual robotic agents possess localized intelligence and operate without a master controller directing task assignments. The swarm relies on a shared global environment visualization state, allowing each agent to independently perform localization, path estimation, behavioral logic adjustment, and conflict resolution.

Engineering Objective: To deliver a scalable framework where individual units can enter or exit the operational environment dynamically without disrupting ongoing material transport arrays.

1.3 Structural Data Flow Loop

The technical pipeline functions across a high-speed closed loop connecting physical spaces to digital mapping canvases:

- **Global Vision Acquisition:** An overhead digital video frame captures physical arena movements.
- **Digital Twin Synthesis:** A PyCharm station running OpenCV extracts ArUco matrix IDs, scaling real spatial markers into explicit metric centimeters.
- **Decentralized Network Broadcast:** The station packs coordinates into a unified JSON string format and fires it via UDP to a network broadcast target.
- **Edge Computation:** An onboard ESP32 microcontroller processes the packet stream, executing spatial trigonometry locally to guide the physical motor chassis.

CHAPTER 2

2.1 Working Principle

The system's tracking loop is driven by the real-time detection of top-mounted ArUco markers. These square, binary fiducial markers consist of a thick black outer border enclosing an inner binary matrix that encodes a unique identifier. The OpenCV ArUco Detector algorithm scans each video frame to identify the outer contours, performs perspective checks to verify the square boundary, and decodes the internal grid against the DICT_4X4_100 matrix dictionary.

- Once a valid marker is identified, the algorithm returns the 2D pixel coordinates of its four corners.
- The marker's centroid (C_x , C_y) is calculated as the mean of its four corner coordinates.
- The absolute heading angle θ of the robot in the global coordinate frame is computed by evaluating the arc tangent of the vector from the bottom-left corner p_1 to the top-left corner p_2 .
- This visual tracking loop provides an absolute, real-time spatial coordinate and heading reference for each active robot.

2.2 Algorithms / Techniques

Algorithms / Techniques used the viability of a decentralized swarm relies on the precision of the shared spatial reality. This reality is synthesized by a PyCharm-based management dashboard operating an OpenCV Python pipeline, which acts as the visual and calculation core of the indoor localization grid.

2.3 ArUco Marker Detection and Pose Estimation

Fiducial markers are binary square patterns utilized for robust robotic localization and augmented reality tracking. The system employs the DICT_4X4_100 ArUco dictionary, a predefined matrix library that offers an optimal balance between processing speed, detection reliability, and physical compactness.

The OpenCV ArUco Detector algorithm scans the raw video frames to identify the black borders and decodes the internal binary grid to return the marker's unique ID and its 2D corner positions. By calculating the mean coordinates of the four corners, the sub-pixel centroid (C_x , C_y) is derived. Furthermore, the absolute orientation angle in radians is

extracted using the arc tangent function on the directional vector formed by the bottom and top corners. This process guarantees that every agent's orientation and position are continuously updated.

2.4 Perspective Projection and Homography

Raw pixel coordinates do not possess true dimensional significance for Mechatronic navigation. Translating these pixels into physical centimeters across a 120 cm x 120 cm arena requires rigorous spatial transformation metrics. While simple linear scaling (Scale_X and Scale_Y) suffices for perfectly parallel camera alignments, real-world camera installations inherently introduce perspective distortion.

Homography, or projective transformation, provides a geometric framework that preserves straight lines while mapping points from one plane (the warehouse floor) to another (the camera's image sensor). If points p_1, p_2, p_3 are collinear in the real world, their transformed counterparts p'_1, p'_2, p'_3 remain collinear in the image plane. The homography function $p' = H p$ is represented in homogeneous coordinates $[x, y, w]^T$ by multiplying the input vector by a 3×3 transformation matrix H : Because the scale factor is arbitrary in homogeneous coordinates, the matrix H possesses exactly 8 degrees of freedom.

2.4.1 Direct Linear Transform (DLT) for Homography Estimation

To correct camera perspective and lens distortion, we implement projective homography to map coordinate planes. A homography is an invertible projective transformation that relates coplanar points between two views.

In homogeneous coordinates, this projection is modeled as: Converting homogeneous projections back to heterogeneous coordinates yields: The homography matrix H has 9 elements but operates with 8 degrees of freedom, as it is defined up to an arbitrary scale factor.

To estimate H , we implement the Direct Linear Transform (DLT) algorithm, which requires a minimum of four coplanar point correspondences. For each corresponding point pair $(x_i, y_i) \rightarrow (x'_i, y'_i)$, we construct a pair of homogeneous linear equations:

Rearranging these equations into a matrix system yields:

where A_i is a 2×9 coefficient matrix: and h is a 9×1 vector containing the elements of H : By stacking the coefficient matrices A_i for all N correspondences, we assemble a global measurement matrix A of size $2N \times 9$: To compute the optimal, non-trivial solution for h that minimizes the algebraic error $\|Ah\|^2$ subject to the unit norm constraint $\|h\| = 1$, we apply Singular Value Decomposition (SVD) to A : The singular values in the

diagonal matrix D are sorted in descending order. The optimal least-squares solution for the vector h corresponds to the last column of the matrix V , associated with the smallest singular value in D . This vector is reshaped into the 3×3 homography matrix H , which is utilized to scale raw camera pixels into physical centimeters across the 120 cm x 120 cm test track.

2.4.2 Differential Drive Kinematics

Each robot features a differential-drive locomotion configuration, controlled by independently regulating the speeds of its left and right wheels. Let (x, y) denote the robot's centroid position and θ denote its orientation heading in the global coordinate frame. The non-holonomic constraint of pure rolling without lateral skidding is modeled as: Given wheel radius R , track width L , and angular velocities of the right and left wheels (ω_R and ω_L), the forward linear velocity V and angular velocity ω are calculated as:

The state-space model for differential drive kinematics is represented as: During navigation, the robot must rotate about its Instantaneous Center of Curvature (ICC). The signed distance $R_{\{ICC\}}$ from the robot's midpoint to the ICC along the common wheel axis is derived as:

To guide the robot to a target coordinate, the edge processor solves the inverse kinematics problem, translating desired linear velocity V and angular velocity ω into target wheel speeds:

2.4.3 Principle of Virtual Work and Scissor Linkage Mechanics

The robotic gripper employs a multi-stage scissor linkage to manipulate payloads. Scissor mechanisms convert linear input displacement into amplified vertical/parallel output motion. Analyzing the actuator forces required to secure a payload involves the principles of Statics and Virtual Work. The principle of virtual work states that for a system of connected rigid bodies in equilibrium, the total virtual work done by all active external forces during any arbitrary infinitesimal virtual displacement is zero:

For a scissor mechanism with N sequential stages, the force required from the linear actuator is highly non-linear and exhibits two distinct scaling behaviors:

1. **Linear Scaling (Payload):** The force required to lift or clamp a payload ($W_{\{load\}}$) sits at the top of the stack and scales linearly with the number of stages N :
2. **Quadratic Scaling (Structural Self-Weight):** The weight of the scissor arms ($W_{\{arm\}}$) is distributed throughout the mechanism. The bottom stage must lift and

support the cumulative weight of all subsequent stages above it, causing the required actuation force to scale quadratically with the stages:

By establishing the static equilibrium equations using virtual work , the total linear force F_{total} required from the actuator at a given scissor linkage angle θ is derived as:

This derivation highlights a critical Mechatronic constraint known as the "Crunch Zone." As the mechanism folds flat, the arm angle θ approaches 0° . Since $\tan(0) = 0$, the theoretical force required to initiate movement from a completely flat state approaches infinity , which would instantly stall the SG90 micro servo. To prevent mechanical stalls and servo damage, our design and firmware physically restrict the gripper's travel to a minimum angle of 15° , ensuring it operates within an efficient mechanical advantage window.

2.5 Data Collection / Data Acquisition

Data is acquired continuously in real-time. An IP webcam streams high-resolution video frames over a dedicated local Wi-Fi router to the Python digital twin. The Python OpenCV script identifies the ArUco marker centroids and headings, applies the homography matrix to convert pixel positions to metric centimeter coordinates , and broadcasts the structured state data across the local network at a frequency of 30Hz.

2.6 Python Digital Twin & Tracking Dashboard

The management dashboard forms the visual and calculation core of the indoor localization grid. Operating inside a Python environment using OpenCV, it transforms raw image tracking feeds into calibrated metric data loops.

2.7 Homography & Real-to-Pixel Coordinate Scaling

To bridge the physical test track to virtual software frames, the system implements linear spatial transformation metrics. Pixels do not hold true meaning for Mechatronic navigation; hence, a conversion constant translates coordinates to physical centimeters based on a defined 120 cm x 120 cm arena dimension.

Scale Factor Equations:

$$\text{Scale_X} = \text{Window_Width} / \text{Arena_Width_CM}$$

$$\text{Scale_Y} = \text{Window_Height} / \text{Arena_Height_CM}$$

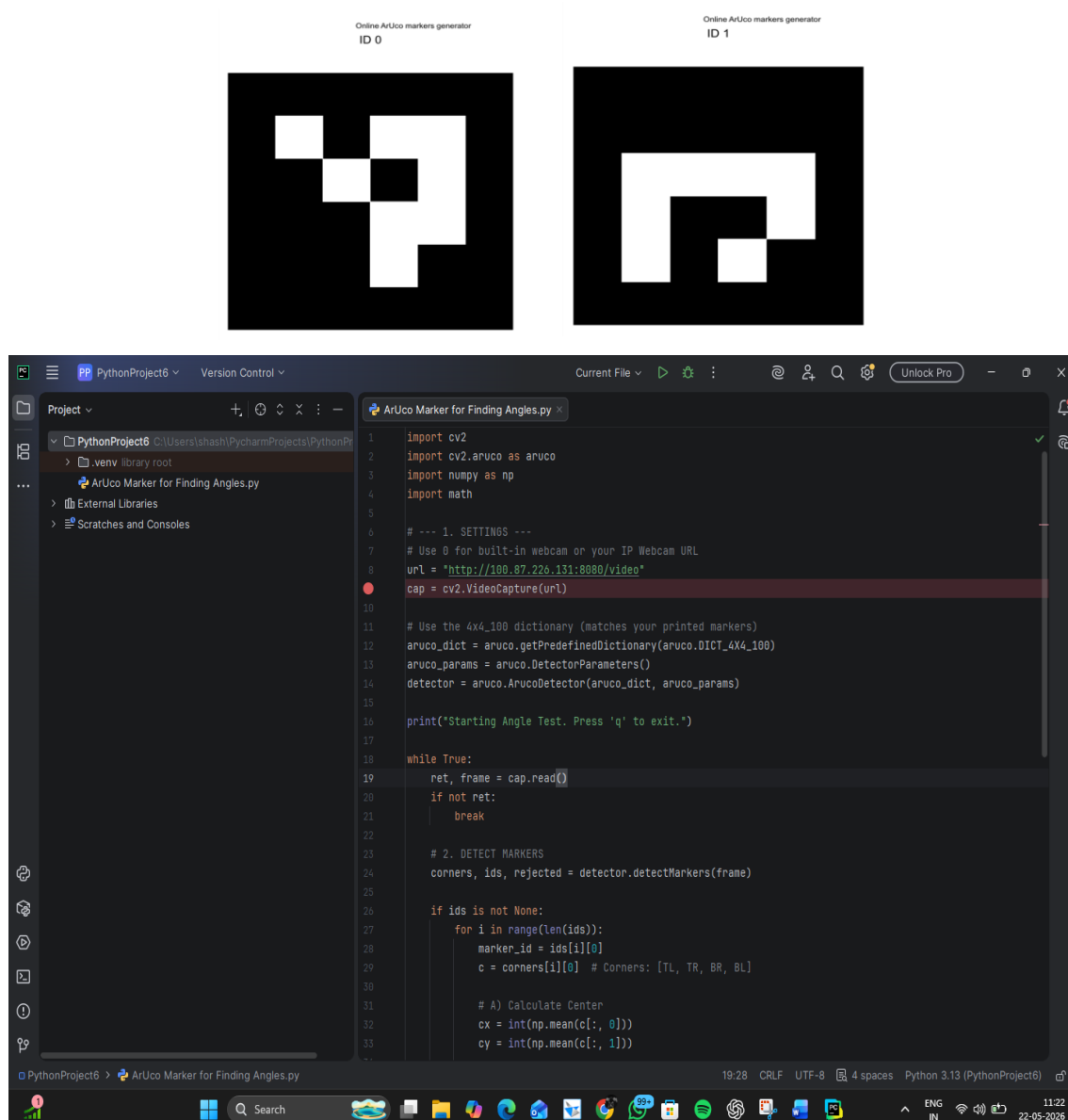


Figure 2.1: Online ArUco Tracking Token Matrices (ID 0 and ID 1)

2.8 Comprehensive Core Dashboard Engine Script

The Python code snippet below demonstrates the video scaling canvas configuration, ArUco identification matrix, color-coded tracking dictionaries, and live UDP JSON stream delivery:

```
import cv2
import cv2.aruco as aruco
import numpy as np
import socket
import json
import math

ARENA_WIDTH_CM = 120.0
ARENA_HEIGHT_CM = 120.0
```

```

DISPLAY_WIDTH = 800
DISPLAY_HEIGHT = 800

SCALE_X = DISPLAY_WIDTH / ARENA_WIDTH_CM
SCALE_Y = DISPLAY_HEIGHT / ARENA_HEIGHT_CM

def cm_to_px(x_cm, y_cm):
    return int(x_cm * SCALE_X), int(y_cm * SCALE_Y)

UDP_IP = "255.255.255.255"
UDP_PORT = 5005
sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
sock.setsockopt(socket.SOL_SOCKET, socket.SO_BROADCAST, 1)

STATUS_COLORS = {
    "IDLE": (0, 255, 0),    # Green
    "BUSY": (0, 255, 255),  # Yellow
    "PICKING": (255, 0, 255), # Magenta
    "ERROR": (0, 0, 255)    # Red
}

bot_states = {0: "IDLE", 1: "BUSY"}
cap = cv2.VideoCapture("http://192.168.1.5:8080/video")

aruco_dict = aruco.getPredefinedDictionary(aruco.DICT_4X4_100)
aruco_params = aruco.DetectorParameters()
detector = aruco.ArucoDetector(aruco_dict, aruco_params)

def draw_scaled_warehouse(ui_frame):
    for i in range(1, 4):
        x_pos = int(i * (DISPLAY_WIDTH / 4))
        y_pos = int(i * (DISPLAY_HEIGHT / 4))
        cv2.line(ui_frame, (x_pos, 0), (x_pos, DISPLAY_HEIGHT), (60, 60, 60), 1)
        cv2.line(ui_frame, (0, y_pos), (DISPLAY_WIDTH, y_pos), (60, 60, 60), 1)

while True:
    ret, frame = cap.read()
    if not ret: break
    frame = cv2.resize(frame, (DISPLAY_WIDTH, DISPLAY_HEIGHT))
    dashboard = np.zeros((DISPLAY_HEIGHT, DISPLAY_WIDTH, 3), dtype=np.uint8)
    draw_scaled_warehouse(dashboard)

    corners, ids, rejected = detector.detectMarkers(frame)
    swarm_data = {}

    if ids is not None:
        for i in range(len(ids)):
            marker_id = int(ids[i][0])
            c = corners[i][0]
            cx, cy = int(np.mean(c[:, 0])), int(np.mean(c[:, 1]))
            cx_cm, cy_cm = round(cx / SCALE_X, 1), round(cy / SCALE_Y, 1)

            angle_rad = math.atan2(c[1][1] - c[0][1], c[1][0] - c[0][0])
            angle_deg = round(math.degrees(angle_rad), 1)
            current_status = bot_states.get(marker_id, "IDLE")
            bot_color = STATUS_COLORS.get(current_status, (255, 255, 255))

            cv2.circle(dashboard, (cx, cy), 25, bot_color, 3)
            swarm_data[f'id{marker_id}'] = {"x": cx_cm, "y": cy_cm, "ang": angle_deg, "status":
current_status}

```

```

if swarm_data:
    sock.sendto(json.dumps(swarm_data).encode(), (UDP_IP, UDP_PORT))
cv2.imshow("Real-Time Metric Swarm Dashboard", dashboard)
if cv2.waitKey(1) & 0xFF == ord('q'): break
cap.release()
cv2.destroyAllWindows()

```

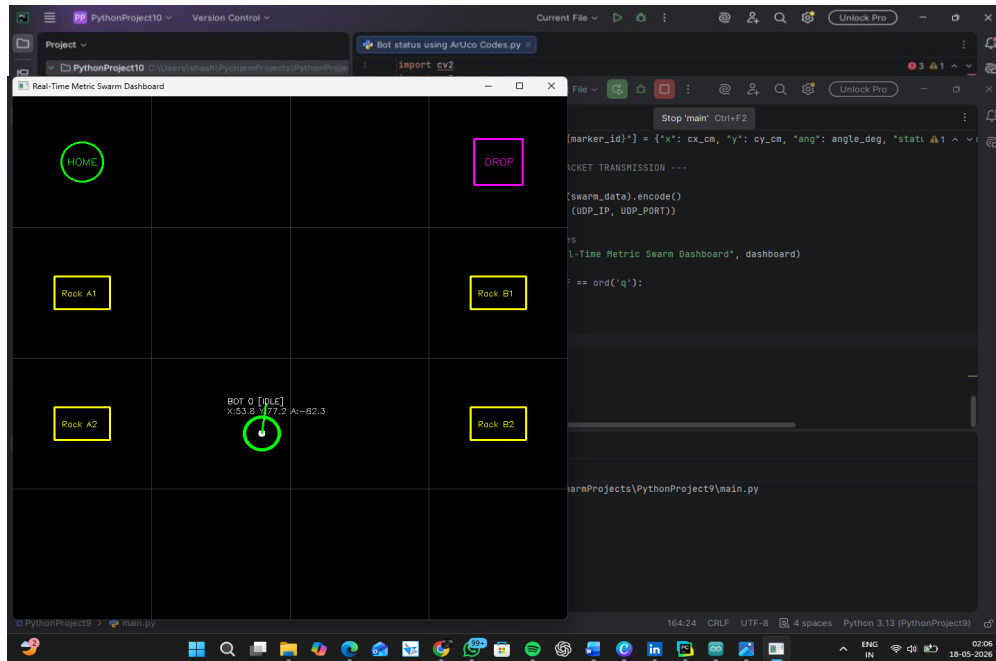


Figure 2.2: Live Visual Representation of the Real-Time Metric Swarm Dashboard

CHAPTER 3

3.1 Mechanical Design & Structural Body Modeling

The physical structure of the swarm agent is designed using a multi-tiered platform architecture. This layout systematically isolates heavy electromechanical assemblies from computational logic elements to maintain high operational integrity.

3.2 Tiered Chassis Topology

The mobile agent's body is divided into two distinct primary layers managed through custom geometric layouts:

- **Lower Deck (LD):** Serves as the structural power platform. It houses the high-current 12V 100RPM DC metal gear motors, heavy-duty differential wheel assemblies, and the core 7.4V Lithium-Ion power source pack.
- **Upper Deck (UD):** Acts as the clean digital acquisition deck. It maintains a completely flat plane layout optimized for hosting the overhead vision ArUco tokens, localized sensor arrays, and the ESP32 microcontroller.

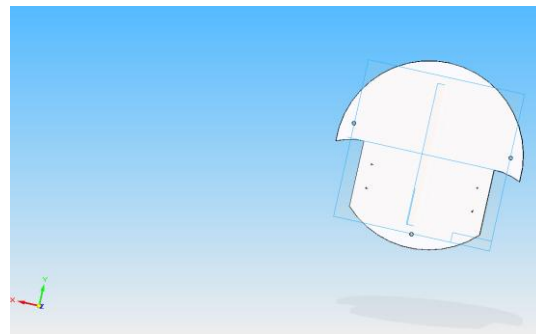


Figure 3.1: 3D CAD Geometric Structural Shell Layout - Lower Deck Profile (LD)

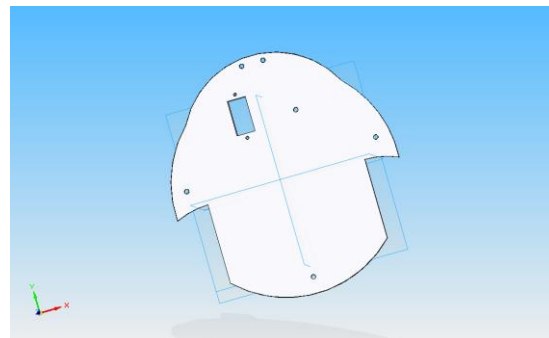


Figure 3.2: 3D CAD Geometric Structural Shell Layout - Upper Deck Profile (UD)

3.3 Material Selection & Manufacturing Feasibility

The plates are modeled and manufactured using high-density lightweight acrylic polymer sheeting. This selection balances rigid structural loading capabilities with effective damping characteristics against high-frequency motor vibration harmonics. It also provides excellent electrical insulation across the chassis plates.

3.4 Mechanical Gripper Linkage Geometry

To execute physical inventory handling, the front-end features a custom gear-driven scissor linkage claw mechanism. Powered by an SG90 servo motor, it provides a highly repeatable operational clearance sweep moving from a 180-degree wide initialization state to a 100-degree mechanical locking configuration to grip package boxes safely.

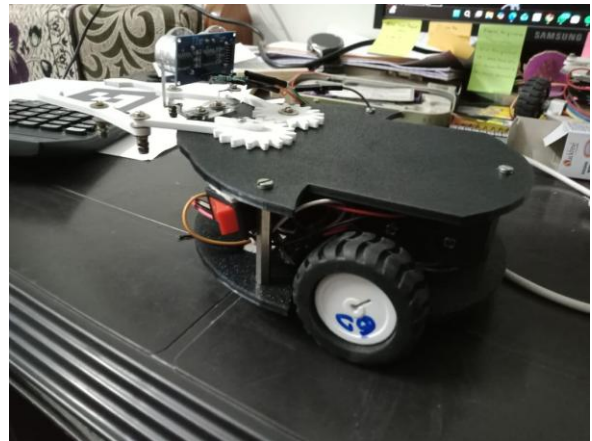
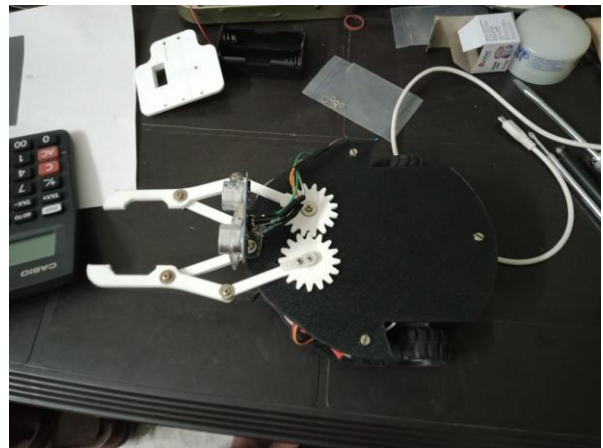


Figure 3.3: Physical Prototyping Assembly & End-Effector Scissor Gripper Mechanism Close-ups

CHAPTER 4

4.1 Electrical Architecture & Hardware Integration

The electrical control distribution routes signals systematically to optimize high-frequency data isolation and prevent internal circuit sags.

4.2 Core Pin Connection Profile

The control hardware centers around a 30-pin ESP32 DevKit V1 module routed to a dual H-bridge TB6612FNG motor driver using a general-purpose copper dot-board perfboard prototyping panel.

Target Component	Pin Label	ESP32 Target GPIO Mapping	Functional System Purpose
TB6612FNG Driver	STBY	GPIO 17 (TX2)	Standby Override Link (Kept HIGH)
TB6612FNG Driver	AIN1	GPIO 18 (D18)	Left Motor Input Directional Control Pin 1
TB6612FNG Driver	AIN2	GPIO 19 (D19)	Left Motor Input Directional Control Pin 2
TB6612FNG Driver	PWMA	GPIO 5 (D5)	Left Motor Speed Vector (Pulse Width Mod)
TB6612FNG Driver	BIN1	GPIO 21 (D21)	Right Motor Input Directional Control Pin 1
TB6612FNG Driver	BIN2	GPIO 22 (D22)	Right Motor Input Directional Control Pin 2
TB6612FNG Driver	PWMB	GPIO 23 (D23)	Right Motor Speed Vector (Pulse Width Mod)
HC-SR04 Ultrasonic	TRIG	GPIO 27 (D27)	Safe Trigger Wave Emission IO Channel
HC-SR04 Ultrasonic	ECHO	GPIO 26 (D26)	Safe Echo Capture Bounce Return Pulse Channel

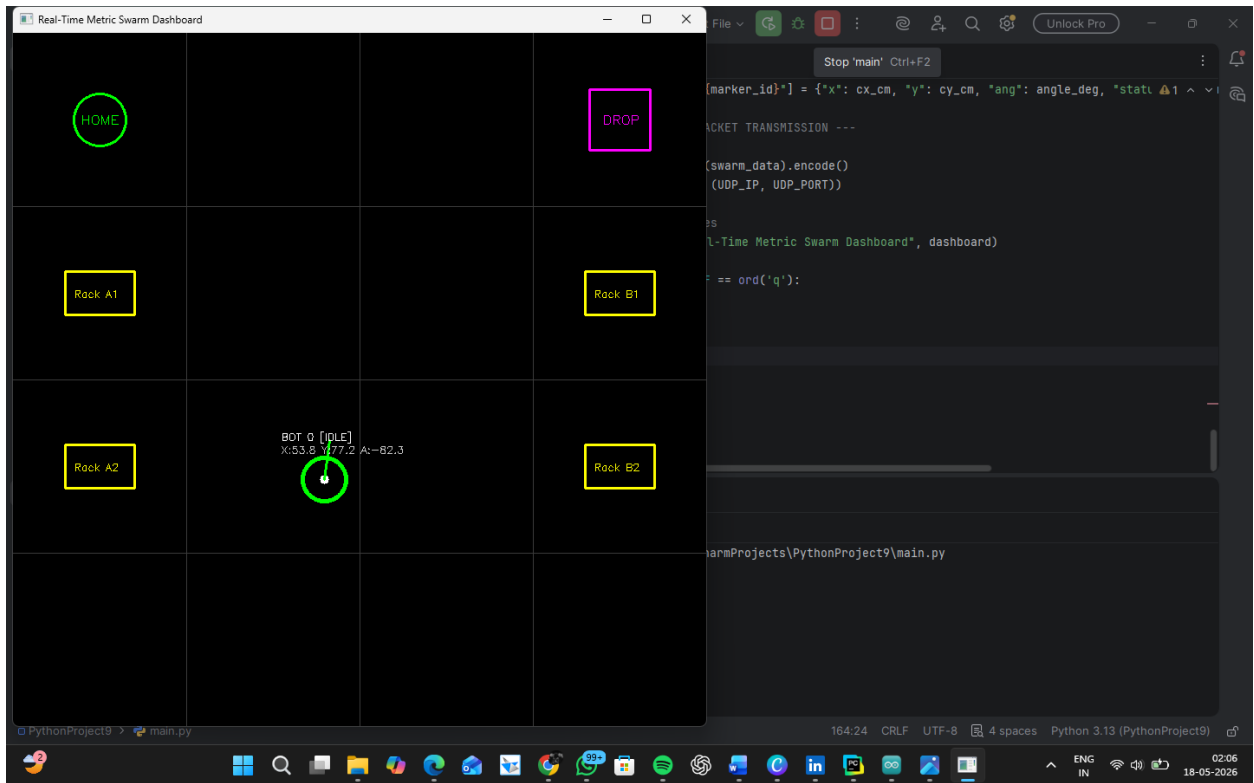


Figure 4.1: Schematic Block Diagram of the Whole Integrated Circuit Distribution System

4.3 Motor Driver Power Split Rules (VM vs VCC)

To protect against electrical 'noise' sags and unexpected brownouts, power routing is handled through separate tracks:

- **VM (Motor Power):** Tied straight to the raw un-regulated 7.4V battery loop to handle high inductive current surges from the wheels.
- **VCC (Logic Power):** Connected to the stable 5V output of the ESP32 onboard regulator to keep processing gates isolated from motor interference.

CHAPTER 5

5.1 System Development

System development was split into three parallel workflows: mechatronic fabrication, host dashboard development, and embedded firmware programming. The physical chassis plates were laser-cut from acrylic sheets, and the electrical circuits were soldered onto prototype perfboards. The host dashboard and coordinate scaling algorithm were written in Python, while the edge navigation logic was implemented in C++ and flashed to the ESP32 units.

5.2 Module Description

The system is structured as three primary functional modules:

1. **Host Dashboard Module (Python):** Acquires raw camera frames, detects the ArUco marker centroids and orientations, applies the homography matrix, and broadcasts a unified JSON packet stream to the network.
2. **Edge Communication Module (ESP32):** Connects to the local network, opens a passive UDP port, and parses the JSON broadcast stream to extract the agent's relative coordinates.
3. **Embedded Navigation & Control Module (ESP32):** Computes target heading vectors, executes proportional differential speed adjustments, and processes localized sensor interrupts to actuate the gripper.

5.3 Integration of Components

The ESP32 DevKit is interfaced with the TB6612FNG driver and peripheral sensors using structured GPIO pin mapping:

- **Standby Pin (STBY):** Tied to GPIO 17 (TX2) and driven HIGH to enable the driver IC.
- **Left Motor Inputs (AIN1/AIN2):** Interfaced with GPIO 18 and 19 to establish H-bridge rotational polarity (forward/reverse).
- **Left Motor PWM (PWMA):** Connected to GPIO 5 to regulate motor speed via hardware-backed PWM.
- **Right Motor Inputs (BIN1/BIN2):** Routed to GPIO 21 and 22 to control right motor rotation direction.
- **Right Motor PWM (PWMB):** Connected to GPIO 23 to regulate right motor speed.

- **Ultrasonic Sensor (TRIG/ECHO):** Interfaced with GPIO 27 and 26 to trigger and receive proximity pulses.
- **SG90 Servo Signal:** Routed to GPIO 13 to control the gripper sweep using a 50Hz PWM signal.

5.4 Fabrication / Assembly Process

The physical robot is constructed using a tiered dual-deck architecture to isolate components and maintain high mechatronic stability. The **Lower Deck** acts as the power stage, housing the high-current 12V DC gear motors, heavy-duty differential wheels, and the 7.4V battery pack. This keeps the robot's center of gravity low, reducing physical swaying during rapid rotational shifts.

The Upper Deck serves as the clean digital acquisition plane, hosting the top-mounted ArUco marker (ensuring unobstructed overhead visibility), the ESP32 microcontroller, and the HC-SR04 ultrasonic array.

5.5 Onboard Decentralized Swarm Agent Firmware

The edge processing firmware running locally on each independent ESP32 unit establishes network connection handshakes, deserializes incoming matrix packets via parsing allocations, and operates absolute trigonometry functions to steer mechanical wheels.

5.6 Onboard Embedded Agent C++ Engine Code

```
#include <WiFi.h>
#include <WiFiUdp.h>
#include <ArduinoJson.h>
#include <ESP32Servo.h>

const char* ssid = "Redmi";
const char* password = "s1s2c7.v";
const int UDP_PORT = 5005;
WiFiUDP udp;

const int MY_ID = 0;
const int CLAW_OPEN = 180;
const int CLAW_CLOSE = 100;

float targetX = 15.0; float targetY = 45.0;
const int AIN1 = 18; const int AIN2 = 19; const int PWMA = 5;
const int BIN1 = 21; const int BIN2 = 22; const int PWMB = 23;
const int STBY = 17;

Servo clawServo;
const int SERVO_PIN = 13;
const int TRIG_PIN = 27; const int ECHO_PIN = 26;
```

```

void drive(int leftSpeed, int rightSpeed) {
    digitalWrite(AIN1, leftSpeed > 0 ? HIGH : LOW);
    digitalWrite(AIN2, leftSpeed > 0 ? LOW : HIGH);
    analogWrite(PWMA, abs(leftSpeed));
    digitalWrite(BIN1, rightSpeed > 0 ? HIGH : LOW);
    digitalWrite(BIN2, rightSpeed > 0 ? LOW : HIGH);
    analogWrite(PWMB, abs(rightSpeed));
}

void setup() {
    Serial.begin(115200);
    pinMode(AIN1, OUTPUT); pinMode(AIN2, OUTPUT); pinMode(PWMA, OUTPUT);
    pinMode(BIN1, OUTPUT); pinMode(BIN2, OUTPUT); pinMode(PWMB, OUTPUT);
    pinMode(STBY, OUTPUT); digitalWrite(STBY, HIGH);
    WiFi.begin(ssid, password);
    while (WiFi.status() != WL_CONNECTED) { delay(500); }
    udp.begin(UDP_PORT);
}

void loop() {
    int packetSize = udp.parsePacket();
    if (packetSize) {
        char packetBuffer[512];
        int len = udp.read(packetBuffer, 511);
        if (len > 0) packetBuffer[len] = 0;
        JsonDocument doc;
        deserializeJson(doc, packetBuffer);
        String myKey = "id" + String(MY_ID);
        if (doc.containsKey(myKey)) {
            float currentX = doc[myKey]["x"];
            float currentY = doc[myKey]["y"];
            float currentAng = doc[myKey]["ang"];

            float dx = targetX - currentX;
            float dy = targetY - currentY;
            float distance = sqrt(dx*dx + dy*dy);

            if (distance < 12.0) {
                drive(0,0); // Reach Target sequence
                return;
            }

            float targetAng = atan2(dy, dx) * 180.0 / M_PI;
            float error = targetAng - currentAng;
            while (error > 180) error -= 360;
            while (error < -180) error += 360;

            if (abs(error) > 25.0) {
                if (error > 0) drive(110, -110);
                else drive(-110, 110);
            } else {
                drive(140, 140);
            }
        }
    }
}

```

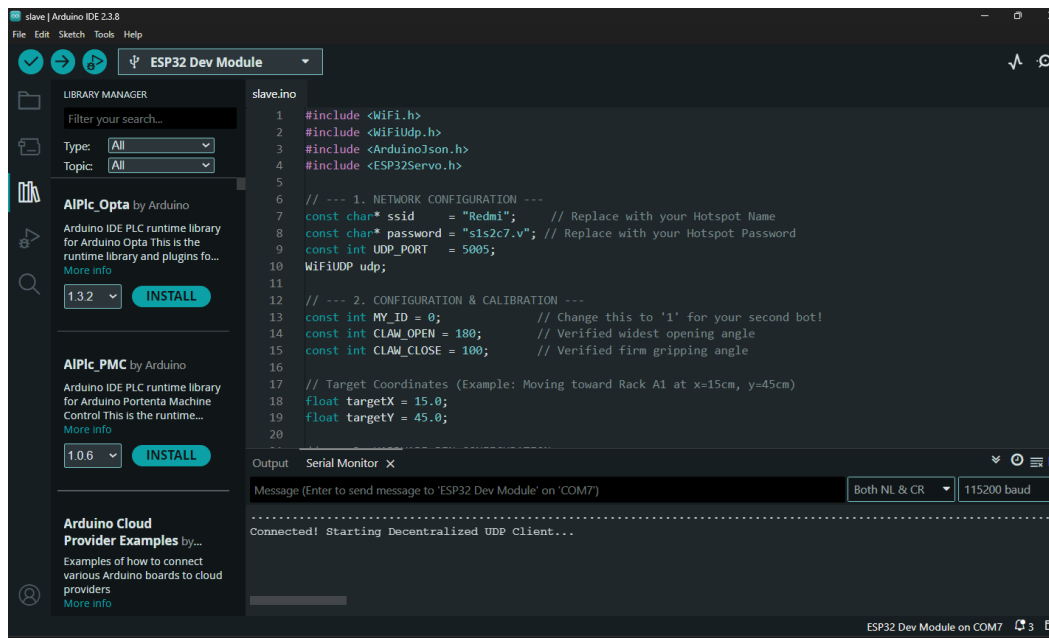


Figure 5.6: Onboard Embedded Agent C++ Engine Code

5.7 System Verification & Presentation Benchmarks

During functional evaluations, the integrated mechatronic system successfully achieved the following project deliverables:

- **Proportional Trajectory Error Realignment:** The agent parses the global map matrix, computing deviation vectors and spinning on its center axis to correct its alignment whenever the angle error exceeds 25 degrees.
- **Tactile Sensor Fusion Handoff:** The moment the HC-SR04 ultrasonic echo registers a local proximity of 4 cm or less, the system interrupts the trajectory path to activate the servo-driven claw, securing the payload firmly between its calibrated 180-degree open and 100-degree closed mechanical limits.

Network Infrastructure, UDP Broadcast, and Embedded Firmware the realization of a decentralized architecture relies fundamentally on a high-speed, scalable network backbone. The Python digital twin continuously extracts the coordinates of all agents and packages them into a unified JSON string. Rather than managing individual TCP connections for every robot, which introduces massive overhead and serialization delays, the system employs a UDP network broadcast.

5.8 UDP Broadcast Architecture in ESP32 Networks

The User Datagram Protocol (UDP) is a connectionless, "fire-and-forget" communication standard. It eliminates the heavy handshakes, flow control, and error correction inherent in TCP, resulting in vastly superior latency. The Python script transmits the JSON payload to the universal broadcast IP 255.255.255.255 on port 5005.

This architecture provides immense scalability; whether the swarm consists of five agents or five hundred, the central transmitter's workload remains identical. Each independent ESP32 edge node passively listens to the network, parsing the packet stream and extracting only the data dictionary associated with its hardcoded identifier (e.g., "id0").

5.9 Packet Loss Benchmarks, Latency Optimization, and Power Save Mitigation

While UDP multicast over Wi-Fi offers unparalleled scalability, it suffers from significant reliability issues, with some dense RF environments experiencing packet loss ratios exceeding 90%. This degradation is not a failure of the microcontroller hardware, but rather an interaction between the UDP protocol and the IEEE 802.11 Medium Access Control (MAC) layer.

A primary cause of massive packet loss is the IEEE 802.11 Power Save mechanism enabled by default on many mobile devices and microcontrollers. When the Wi-Fi modem momentarily sleeps to conserve power, it misses unbuffered broadcast frames. While an Access Point (AP) will buffer unicast packets for sleeping clients, it does not guarantee the buffering of broadcast packets. Furthermore, broadcast packets are transmitted at lower base data rates without MAC-level acknowledgments (ACKs), making them highly susceptible to CSMA/CA (Carrier-Sense Multiple Access with Collision Avoidance) contention and RF interference.

To mitigate these latency and packet drop issues, several aggressive firmware optimizations are executed within the ESP-IDF framework :

- **Dynamic Buffer Sizing:** The memory allocation for the Lightweight IP (LwIP) and Wi-Fi drivers is carefully managed. By increasing the dynamic RX buffers (CONFIG_ESP32_WIFI_DYNAMIC_RX_BUFFER_NUM), the microcontroller can absorb rapid bursts of packets during network jitter without suffering from buffer underruns.
- **API Optimization:** Inefficient looping paradigms—such as repeatedly invoking

beginPacket() and endPacket() routines inside the main loop—are eliminated to prevent socket exhaustion and artificial transmission delays.

- **High-Frequency Redundancy:** Recognizing that UDP guarantees no delivery, the system relies on sheer volume. Because the Python tracking engine operates at standard video framerates (~30 frames per second), the dashboard broadcasts coordinate updates roughly every 33 milliseconds. Even if 50% of the packets are lost due to RF collisions, the robot receives fresh trajectory data every 66 milliseconds. In the context of the physical inertia of a moving robot, this fraction of a second delay is mechanically imperceptible.

5.10 Onboard Embedded Agent Firmware and Local Edge Computation

The edge processing firmware running locally on each independent ESP32 unit is responsible for establishing the Wi-Fi connection handshake, initializing the UDP listener on port 5005, and managing the electromechanical hardware.

When a packet is parsed successfully, it is loaded into a character buffer. The ArduinoJson library is invoked to deserialize the JSON document. The firmware checks for the presence of the agent's unique key (`myKey = "id" + String(MY_ID)`). If the key exists, it extracts the current X coordinate, Y coordinate, and angular heading (`\theta`).

The agent continuously calculates the spatial deviation to its assigned target destination (`target_X`, `target_Y`) :

Subsequently, the exact angular heading required to intercept the target is derived using the 4-quadrant arctangent function:

By computing the deviation between the target angle and its current absolute orientation (`error = target\_Angle - current\_Angle`), the ESP32 calculates the precise differential drive vector. The firmware programmatically bounds this angular error between -180 and 180 degrees using nested while loops to ensure the robot always rotates via the shortest possible arc, optimizing energy consumption and transit time.

Chapter 6:

System Verification, Presentation Benchmarks, and Conclusion

During the Phase 1 Continuous Internal Evaluation (CIE), the integrated mechatronic system was subjected to rigorous functional benchmarking to validate the efficacy of the decentralized architecture. The testing confirmed that shifting the computational path planning and error correction burdens from a central server to independent edge nodes provides immense operational advantages.

6.1 Proportional Trajectory Error Realignment

The physical agents demonstrated highly responsive, real-time path correction capabilities. When traversing the arena, the local ESP32 continuously evaluates the error vector parsed from the global map matrix. The logic gates enforce a strict proportional realignment parameter: whenever the angular error exceeds ± 25 degrees, the agent immediately halts forward momentum. The TB6612FNG motor driver is commanded to invert the polarity of one wheel while maintaining the other, spinning the agent perfectly on its center axis (ICC). Once the orientation error falls below the 25-degree threshold, the H-bridge normalizes, applying a synchronized PWM signal to both motors to drive the chassis forward aggressively. This logic sequence successfully bridges the gap between theoretical kinematics and physical mechanical inertia, ensuring stable, oscillation-free transit.

6.2 Tactile Sensor Fusion Handoff

While the global ArUco vision tracking provides absolute metric localization, overhead computer vision systems inherently lack the millimeter-level precision and depth perception required for intricate physical manipulation tasks. To circumvent this limitation, the system architecture employs localized sensor fusion.

The ESP32 interfaces with an onboard HC-SR04 ultrasonic echo sensor, emitting high-frequency acoustic pulses and measuring the return bounce to calculate precise proximity distances. The moment the ultrasonic sensor registers a local proximity of 4.0 cm or less to a target payload, the edge generates a hardware interrupt. The ESP32 immediately overrides the global trajectory routing, forcing the motor driver to brake the chassis.

Subsequently, the execution thread commands the SG90 servo to sweep from its 180-

degree open state to its 100-degree locked state, deploying the scissor linkage to secure the payload. Following successful gripping, the chassis reverses momentarily before dynamically rerouting to a new coordinate destination. This seamless operational handoff—transitioning from global visual macro-routing to localized tactile micro-execution—epitomizes the efficacy and resilience of decentralized edge-computing.

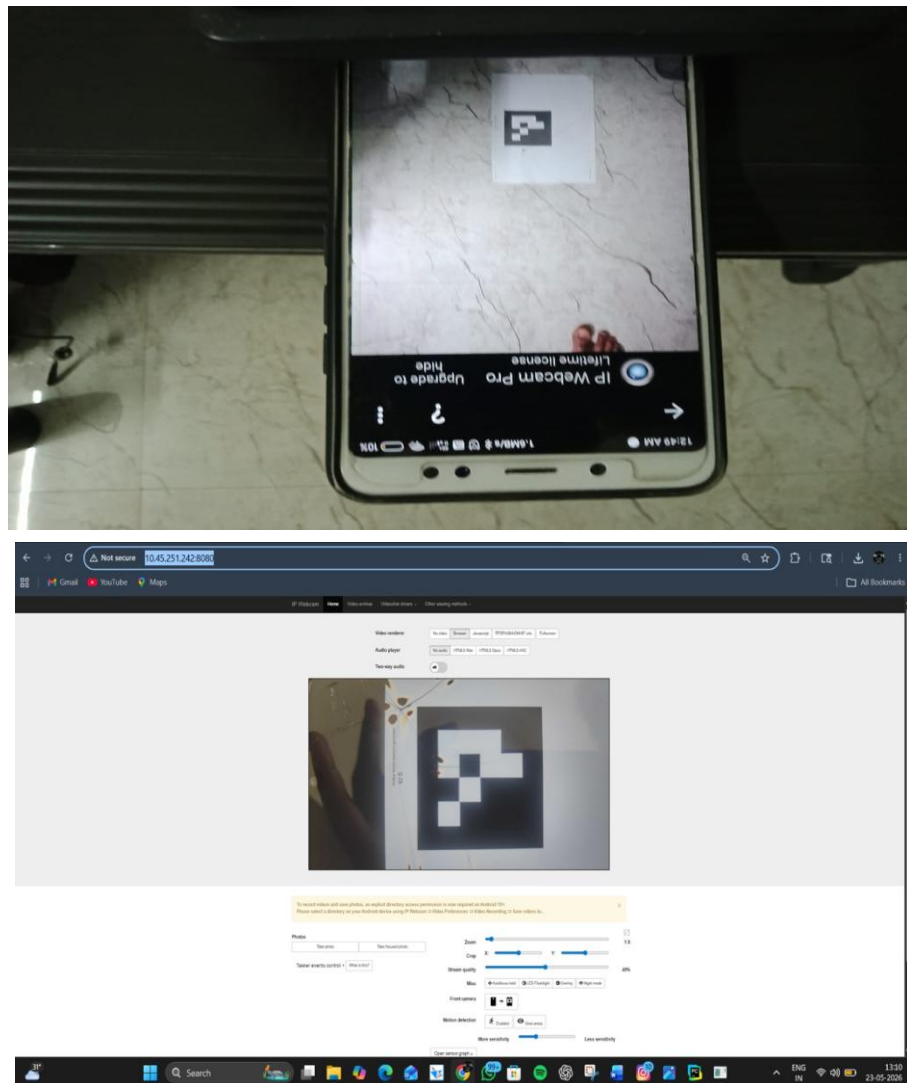


Figure 6.1: System Verification, Presentation Benchmarks

CHAPTER 7

7.1 References

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